Xi'an Jiaotong-Liverpool University
西交利物浦大学

Developing an AI-powered System for Meta-software Development

MARKS: This component is worth 30% of the coursework.

DEADLINE: June 7th, 2026.

WHERE: Electronic submission on Learning Mall.

HOW: Submit a compressed file (zip) named with your student ID and name in the format StudentID-Your_Name.zip. For example, 12345-Netza_Cruz.zip

TASK 1: You should prepare a Jupiter Notebook to **extend a baseline software to deliver an AI-powered system for meta-software development.** Your software should automatically generate code for a functional Flask API and a website. The website should display at least one automatically generated image. Your code should also automatically generate traditional SDLC documentation (such as UML diagrams and requirements appropriate to a given business problem).

TASK 2: You should deploy the Flask API and website generated in TASK 1 and demonstrate your skills by utilising version control and integrating CI/CD practices.

Your code should demonstrate your ability to:

- Select and apply appropriate software development methodologies, demonstrating informed decision-making and alignment with system requirements and constraints.
- Design and modelling of software systems, evidencing the effective use of UML and AI-specific tools to support clarity, structure, and communication of system architectures.
- Integrate CI/CD and AI-specific tools that influence software design choices and modelling practices, including their benefits and limitations.

Learning Outcome	Description
B	Demonstrate the ability to select and apply appropriate software development methodologies
C	Design and Model Software Systems using UML and AI-Specific Tools
D	Implement and Evaluate Testing and Deployment Strategies

Report Requirements

Be mindful to fulfil the requirement listed below:

1. You are free to include libraries into your project, however, there should be only one Jupiter Notebook in your TASK 1.
2. To demonstrate your skills on version control and CI/CD, you **must include three screenshots**, one of your commit records, another of the website deployment, and one showing your CI/CD workflow.
3. Your submission should consist of the below structure:

```
StudentID-Your_Name
|
|_Task1/
|_Task2/
```

ADVICE

To succeed in developing your software, consider using the laboratory sessions. You are welcome to test your code at the university laboratories, utilising the pre-installed conda environment ai_in_se_cw.

REFERENCING SOURCES

You must reference any sources you use. This means that you must include in-text citations and include a complete reference list at the end of your work. For further guidance on the APA Referencing System, please refer to the XJTLU Library guide. Note that referencing should be consistent, accurate, and adhere to XJTLU's Academic Integrity Policy.

ACADEMIC INTEGRITY

Your assignment must be all your own work. Plagiarism, copying, collusion, or dishonest use of data will be penalised. Penalties ranging from capped scores to an award of 0 (zero) will apply. Please refer to the for guidance.

Be advised that more serious instances of academic infringement will be presented at exam boards, which can result in HoDs being notified.

GENERATIVE AI

The written content should be your own work, and GenAI should not be used to create the written content for your visual aid or to prepare a script for your presentation.

STUDY SKILLS

For study skills support, please see the University of Liverpool's KnowHow service. The section on reflective writing is useful for all assignments, and the presentation tutorial is especially relevant for PGC405.

LATENESS

XJTLU policy is -5% per day up to a total of 25%. Work submitted more than five working days late will receive a grade of zero.

Marking Criteria

Task	Criteria	Description	Maximum Marks
Task 1	Application of LLMs and Generative Components in the Solution	Effective implementation of mechanism(s) to generate documentation.	4
		Effectively embed a mechanism(s) to prompt a website with a generated image.	4
		The project (including the notebook) is structured coherently, with informative comments and a professional format.	4
Task 1	Software System Design and Modelling	Effective automatic generation of UML diagram(s).	2
Task 1	Integration of AI-Specific Tooling	Effective use of AI-specific tooling.	3
Task 2	Version Control and Collaboration Practices	Effective use of version control systems.	4
Task 2	Integration of CI/CD	Effective use of workflow.	3
		Effective use of testing practices.	3
		Effective deployment.	3
			30

Appendix 1: Analysis of the Generative AI Usage Policy for the DTS114TC Presentation-Component's Assessment

This policy is designed to be both **permissive** and **protective**. It acknowledges the growing role of AI as a professional tool while safeguarding the module's academic integrity and core learning objectives.

The policy clearly distinguishes between using AI as an **assistant** and using it as a **substitute** for your own intellectual effort.

1. Permitted Uses: AI as a Collaborative Tool

You are encouraged to use Generative AI to *enhance* your work, not create it. The allowed uses align with common tasks where technology can improve efficiency and quality.

- **a) Brainstorming for Inspiration:**
  - **What it means:** You can use AI (e.g., XIPU AI, ChatGPT, DeepSeek, or the module-developed AI agent) to generate initial ideas, mind maps, potential lesson plan structures, or arguments for and against a topic.
  - **Why it's permitted:** This mimics a discussion with a colleague. The final direction, critical selection, and development of ideas must be your own.
- **b) Proof-reading and Language Editing:**
  - **What it means:** You can use AI (e.g., Grammarly, XIPU AI, ChatGPT) specifically to check grammar, spelling, punctuation, and sentence clarity. It can help rephrase awkward sentences *for clarity only*.
- **c) Finding or Generating Images:**
  - **What it means:** You can use AI image generators to create custom illustrations, diagrams, or concept art for your assignment, provided it is relevant.
  - **Why it's permitted:** Visuals can enhance a presentation or report. This is seen as a digital design skill. You must ensure the images are appropriate and correctly referenced if required by the assignment brief.
- **d) Designing Layout:**
  - **What it means:** You can use AI tools to get suggestions on how to structure a document, presentation, or poster for visual appeal and professional presentation.
  - **Why it's permitted:** This focuses on the aesthetics and organisation of information, not the information itself.

2. Prohibited Use: AI as an Author

This is the most critical rule and is non-negotiable.

- **You CANNOT use Generative AI to write the assessments for you.**

- **What it means:** You must not prompt an AI to generate entire paragraphs, sections, or the entire answer to an assessment question and then submit it as your own work. This constitutes academic misconduct (plagiarism).
- **Consequence:** The policy explicitly states that submitted work will be run through AI-writing detection software (e.g., Turnitin AI Detection). If a significant portion of your text is identified as AI-generated, you will likely be investigated for a breach of academic integrity, which can have serious consequences, including failing the module.

Why This Policy Exists

The DTS114TC module aims to develop your **critical thinking, reflective practice, and ability to synthesise theory.**

- **The Process is the Product:** The value of the assessment lies in *your* cognitive process: analysing reading materials, reflecting on your classroom practice, constructing your own arguments, and demonstrating your understanding. Using AI to shortcut this process defeats the entire purpose of the postgraduate certificate.
- **Professional Preparedness:** The policy teaches you to use AI *ethically* and *effectively*, a crucial skill for 21st-century education.
- **Authentic Assessment:** The university needs to verify that *you* have mastered the learning outcomes, not an AI model.

Practical Advice for You as an In-Service Teacher

1. **Always Acknowledge and Reflect:** Even for permitted uses, it is best practice to include a short, honest "**AI Use Statement**" at the end of your assignment.
    - Example: "I used ChatGPT (GPT-4) for brainstorming initial ideas for my pedagogical research topics and for proofreading my final document to correct grammatical errors. All substantive content and critical analysis are my own work."
    - This demonstrates transparency and good academic practice.
2. **You Are the Expert:** Use AI to support and structure your unique insights, not replace them. An AI can give you a generic plan, but it cannot replicate the nuanced understanding you have of your specific context.
3. **Verify Everything:** AI is known for "hallucinating" (making up facts, references, and quotes). Never trust AI-generated information without verifying it against your module's core readings and reputable academic sources.

In summary, this policy encourages you to **use AI as a smart assistant, not a ghostwriter**. Embrace it for enhancing efficiency and presentation, but ensure the intellectual heart of the work, the analysis, reflection, and argument, which is authentically and unmistakably yours.